csdf 1

```

import re

print("===== EMAIL HEADER ANALYZER =====")
print("Paste the raw email header.")
print("Type END on a new line when finished.\n")

lines = []

while True:
    line = input()
    if line.strip() == "END":
        break
    lines.append(line)

header = "\n".join(lines)

# Extract important email fields
fields = [
    "From",
    "To",
    "Subject",
    "Date",
    "Message-ID",
    "Reply-To",
    "Return-Path"
]

print("\n===== HEADER ANALYSIS =====")

for field in fields:
    match = re.search(
        r"(?im)^" + re.escape(field) + r":\s*(.*(?:\n[ \t]+.*)*)",
        header
    )

    if match:
        print(field + ":", " ".join(match.group(1).split()))

# Extract IP addresses
ips = re.findall(
    r"\b(?:\d{1,3}\.){3}\d{1,3}\b",
    header
)

print("\n===== IP ADDRESSES =====")
if ips:
    for ip in dict.fromkeys(ips):
        print(ip)
else:
    print("No IPv4 addresses found.")

# Show email authentication results
print("\n===== AUTHENTICATION =====")

```

```
for check in ["SPF", "DKIM", "DMARC"]:
    results = re.findall(
        r"\b" + check.lower() + r"\s*=\s*
(pass|fail|softfail|neutral|none|temperror|permerror)",
        header,
        re.IGNORECASE
    )

    if results:
        print(check + ":", ", ".join(
            result.upper() for result in results
        ))
    else:
        print(check + ": Not found in header")

# Show Received fields (email delivery path)
print("\n===== EMAIL DELIVERY PATH =====")

received = re.findall(
    r"(?im)^Received:\s*(.*(?:\n[ \t]+.*)*)",
    header
)

if received:
    for i, item in enumerate(received, 1):
        print(str(i) + ".", " ".join(item.split()))
else:
    print("No Received fields found.")

print("\n===== ANALYSIS COMPLETED =====")
print("Note: Results are extracted from the supplied header.")
print("This program does not prove that an email is safe.")
```

```
PS D:\submission\LP4> python -u "d:\submission\LP4\f.py"
Paste your email header below.
Type END on a new line when finished:

From: Internshala <student@updates.internshala.com>
To: spsr@example.com
Date: Wed, 28 Oct 2026 13:33:16 +0000
Subject: Internship Application Update
Message-ID: <demo@example.com>
Reply-To: no-reply-student@internshala.com
Received: from o2649.pnw.mail.internshala.com (159.183.106.129) by mx.google.com with ESMTPS; Wed, 28 Oct 2026 06:33:19 -0700
Authentication-Results: mx.google.com; spf=pass; dkim=pass; dmarc=pass
END

====================================
      EMAIL HEADER ANALYSIS
====================================

EMAIL DETAILS
From: Internshala <student@updates.internshala.com>
To: spsr@example.com
Date: Wed, 28 Oct 2026 13:33:16 +0000
Subject: Internship Application Update
Message-ID: <demo@example.com>
Reply-To: no-reply-student@internshala.com

AUTHENTICATION RESULTS
mx.google.com; spf=pass; dkim=pass; dmarc=pass
SPF: PASS
DKIM: PASS
DMARC: PASS

RECEIVED HOPS (top to bottom)
- from o2649.pnw.mail.internshala.com (159.183.106.129) by mx.google.com with ESMTPS; Wed, 28 Oct 2026 06:33:19 -0700

IP ADDRESSES FOUND
- 159.183.106.[redacted]

NOTE:
Header information can be forged.
Authentication results are not independently verified.
Preserve the original email for investigation.
```

Email Header Analyzer

Trace the origin of any email by analyzing its message headers

Paste Raw Email Headers

Load Sample

```
From: Internshala <student@updates.internshala.com>
To: spsr@gmail.com
Date: Wed, 28 Oct 2026 13:33:16 +0000 (UTC)
Message-ID: <GwCrZzV8*************>
Subject: Update: spsr, important update on your application
Reply-To: no-reply-student@internshala.com
MIME-Version: 1.0
Content-Type: text/html; charset=utf-8
X-Entity-ID: u001.FQMyRn9mrt10Db[redacted]==
```

Your headers stay in your browser. Only the sending server's IP address is sent to look up its location.

Clear

Analyze Headers

Sender IP 159.183.106.129 FOUND 594ms

Copy PDF 3 checks remaining

SENDER IP
159.183.10[redacted]

ORIGIN
San Francisco, California, United States of America

ISP
Twilio SendGrid

ASN
11377

Authentication

SPF	PASS	
DKIM	PASS	updates.internshala.com
DMARC	PASS	internshala.com

As reported by the server that received the message. A pass means the domain allowed this server to send, not that the message is safe.

Delivery Path 2

1 o2649.pnw.mail.internshala.com 159.183.106[redacted] → mx.google.com
ESMTPS SENDING SERVER

2 Created on this server → 2002:a05:6022:3130:b0:119:6bb[redacted]
SMTP +1 sec

Sender Origin Details

IP ADDRESS	159.183.106.[redacted]
COUNTRY	United States of America
REGION / STATE	California
CITY	San Francisco
POSTAL CODE	94105
ISP	Twilio SendGrid
ASN	11377